

Final Project

Transformer-Based NLP Application

1 Overview

- In recent years, Large Language Models (LLMs), with their powerful natural language processing capabilities, have become widely used and valuable tools across a broad range of fields. These models are built upon the Transformer architecture, which represents a major breakthrough in Natural Language Processing (NLP). The self-attention mechanism employed by Transformer models enables them to achieve strong performance in terms of both accuracy and processing speed compared with many earlier architectures.
- In this project, students will study and develop a practical application using a model based on the Transformer architecture.

2 Requirements

- First, students must select a problem in the field of Natural Language Processing. Suggested tasks include:
 - Part-of-Speech (PoS) Tagging.
 - Text Classification.
 - Named Entity Recognition.
 - Machine Translation.
 - Text Summarization.
 - Question Answering.
 - Other NLP tasks.
- Students must then select a model based on the Transformer architecture to address the chosen problem and complete the following two requirements:
 1. **(8 points)** Students must identify a suitable dataset and train or fine-tune the selected model using that dataset. The trained model must then be evaluated using appropriate evaluation metrics. The report must provide a detailed description of the dataset, the selected model, the train, validation, and test data splits, and the model evaluation results based on suitable metrics.
 2. **(2 points)** Students must develop a web-based application using the trained model.

3 Submission Guidelines

- Students must submit the complete source code, trained model weights, training dataset, and project report.
- All submission materials must be packaged into a single ZIP file. The filename must contain the student IDs of all team members, separated by underscores, using the following format:

`StudentID1_StudentID2_..._StudentIDN.zip`

- If the submission file is too large, students may upload it to Google Drive and submit a text file containing the corresponding link. The text file must be named using the following format:

`StudentID1_StudentID2_..._StudentIDN.txt`